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CS-404: Discrete Mathematics Assignment - Topic Introduction.	Name: Mansoor Izzy Enrollment No: 043.

THE EUCLIDEAN ALGORITHMS

★ Divisibility of Numbers

- Let a and b be two integers. We say that a divides b (written as a|b) if there exists an integer m, such that $b = am$.
- If a does not perfectly divide b, we can divide b by a to get a quotient q and an integer remainder r, satisfying the equation
  $b = aq + r$, where $0 \leq r < a$.

- Proving that if c|b and c|a, then c|r:
  - Assume an integer c divides both a and b
  - By definition, $b = cx$ and $a = cy$, for some value x and y.
  - If $b = aq + r$
    $cx = cyq + r$
    or, $r = cx - cyq$
    $r = c(x - yq)$, where $x - yq$ is an integer
    $\therefore c|r$.
- This proof, that any common divisor of two numbers must also divide their remainder is essential to the Euclidean Algorithm.

★ Prime Numbers

- An integer is called a 'prime' $p > 1$, if its only divisors are $1, -1, p$ and $-p$. It cannot be written as the product of two smaller positive integers.

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- Prime factorization: Every positive integer (except 1) that is NOT a prime can be decomposed into a product of prime numbers.
- The Fundamental Theorem of Number Theory states that the prime factorization of any positive integer is unique, regardless of the order of the prime factors.
- Relative Primes or Co-primes are two integers whose greatest common divisor is exactly 1.
  Eg: 4 and 9.

★ Greatest Common Divisor (GCD):

- The greatest common divisor is the largest positive integer that divides two positive integers without a reminder.
  Eg: • Lets take two numbers 6 and 9.
  • The largest positive number that divides both 6 and 9 is 3.
  ∴ GCD (6, 9) = 3.

- Finding GCD using Prime Factorization:
  You can find the GCD by isolating the common prime factors of two numbers and multiplying them together.
  • Eg: Lets take 300 and 18. and their prime factors.
  • $300 = 2^2 \times 3 \times 5^2$, and
  • $18 = 2 \times 3^2$
  • The common factors are 2 and 3
  • gcd (300, 18) = $2 \times 3 = 6$.

- This method is not ideal because it requires you to know the prime factorizations of two numbers. Foreven moderately large numbers, calculating it is extremely difficult and time consuming.

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★ The Euclidean Algorithm

- It is an incredibly fast algorithm for computing the GCD without the need for factorization.
  It was first documented by Greek mathematician Euclid in his book Elements at around 300 BC.

- It works on the mathematical principle that
  $gcd(b, a) = gcd(a, r)$, where $b = aq + r$.
- We proved earlier that any common divisor of b and a also divides r, so we can replace the larger number by the remainder and the greatest divisors will remain.

- Eg: Calculating $gcd(300, 18)$ using the Euclidean algorithm:-
  - $300 = 18 \times 16 + 12$.
  - $\therefore gcd(300, 18) = gcd(18, 12)$
  - $18 = 12 \times 1 + 6$
  - $\therefore gcd(18, 12) = gcd(12, 6)$
  - $12 = 6 \times 2 + 0$
  - $\therefore gcd(12, 6) = gcd(6, 0)$
- We can infer that $gcd(a, 0) = a$ for all values of a.
  $\therefore gcd(300, 18) = 6$.

- Implementation of Euclidean Algorithm can be done using both an iterative approach and a recursive approach.

```
# Iterative Approach
def gcd_iterative(a,b):
    while b!=0:
        rem = a % b
        a = b
        b = rem
    return a.
```

```
# Recursive Approach.
def gcd_recursive(a,b):
    if b==0:
        return a
    return gcd_recursive(b, a%b)
```

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★ Time Complexity Analysis

- The time and steps taken to run the Euclidean algorithm can vary significantly.
- Best case Scenario:
  - Suppose we run the algorithm on $a = n$ and $b = 2n+1$
    - → $2n + 1 = 2 \times n + 1$
    - ① → $\gcd(2n+1, n) = \gcd(n, 1)$
    - → $n = 1 \times n + 0$
    - ② → $\gcd(n, 1) = \gcd(1, 0) = 1$.
  - The algorithm resolves in just 2 steps, despite the fact that $2n+1$ and n are relative primes, share no common factors and can be arbitrarily large.

- Worst Case Scenario:
  - Suppose we run the algorithm on two consequtive Fibonacci Numbers F_{k+1} and F_k.
    - → $F_{k+1} = 1 \times F_k + F_{k-1}$
    - → $F_{k-1} = 1 \times F_{k-1} + F_{k-2}$
    - ⋮
    - → $F_1 = 1 \times F_1 + 0$.
  - The quotient is 1 at every step and the algorithm only terminates after $k-1$ steps.
  - This means the number of steps to run it can be very large if we chose very large numbers on the Fibonacci sequence.

- We know that $b = aq + r$ and since $q \geq 1$,
  $b \geq a + r$.
- We also know that $a > r$, therefore,
  $b > 2r$.
- Multiplying 'a' on both sides,
  $ab > 2ar$ or $\boxed{\frac{ab}{2} > ar}$.

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- This implies that after each step in the algorithm, the product of two working numbers drops by at least a factor of two, they half themselves.
- ∴, the number of execution steps is at most
$$\log_2 a + \log_2 b,$$
resulting in a time complexity of $O(\log_2(a,b))$.

- In 1844, Gabriel Lamé gave the Lamè theorem which formally proved that the Euclidean Algorithm will never take more than 5 times the number of digits in the smaller number.
  - For eg: GCD of two 300-digit numbers takes under 1500 steps.
- This discovery is historically significant as it is considered one of the first practical applications of algorithm analysis.

★ Extended Euclidean Algorithm

- It is an advanced variant of the standard algorithm that not only calculates the GCD, but traces the steps backward to express GCD as a linear equation.
- It is based on Bézout's Identity which states that for any integers a and b, there always exists coefficients x and y, such that
$$ax + by = \gcd(a, b).$$

- Eg: Finding the x, y pair for $\gcd(300, 18) = 6$.
  - working backward from $6 = 18 - 12 \times 1$
  - we have, $12 = 300 - 18 \times 16$
  - Substituting in the previous, $6 = 18 - (300 - 18 \times 6) \times 1$
$$\therefore 6 = 300(-1) + 18(17)$$
here $x = -1$ and $y = 17$ satisfies Bézout's identity.

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★ Applications of Euclidean Algorithms:

I.] Simplify Fractions Instantly:

- By computing the GCD of a fraction's numerator and denominator using the Euclidean Algorithm, we can divide and simplify it instantly.

II.] RSA Cryptography (Rivest- Shamir- Adleman):

- The mathematical foundations of ~~seq~~ secure internet protocalls like HTTPS relies heavily on RSA algorithm.
- In this protocol, two massive (100+ digits) ~~protocols~~ prime numbers p and q are multiplied to get $m = pq$.
- To encrypt data, a public key e is chosen such that it is relatively prime to $(p-1)(q-1)$, a condition that is very rapidly verified by the Eudidean algorithm.
- To generate the ~~seq~~ secure private key d, the extended eudidean algorithm is used to ensure $(p-1)(q-1)$ devides $ed-1$.

III.] Modular Arithmetic:

- The extended Eudidean Algorithm is the standard computational method for finding modular multiplicative inverses.
- These inverses are critical for modern computer algebra systems and are heavily utilized in primality testing algorithms to verify if the large numbers created for cryptography are actually prime or not.